Chapter 5. The Fourier Transform on R

Schwartz space, inversion, Plancherel, Poisson summation

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5.1 Elementary theory of the Fourier transform

Motivation. On $\mathcal{S}(\mathbb{R})$ the transform $\hat{f}(\xi) = \int f(x)e^{-2\pi ix\xi}dx$ swaps differentiation and multiplication by x, fixes the Gaussian, and is an L^2-isometric bijection.

5.1.1 Integration of functions on the real line

f is of **moderate decrease** if continuous with $|f(x)| \le \frac{A}{1+x^2}$; then $\int_{-N}^{N} f$ is Cauchy and $\int_{-\infty}^{\infty} f = \lim_N \int_{-N}^{N} f$. These form a vector space $\mathcal{M}(\mathbb{R})$.

Proposition 1.1

For f, g of moderate decrease:
(i) Linearity: $\int(af + bg) = a\int f + b\int g$.
(ii) Translation invariance: $\int f(x-h)\,dx = \int f(x)\,dx$ for $h \in \mathbb{R}$.
(iii) Scaling: $\delta \int f(\delta x)\,dx = \int f(x)\,dx$ for $\delta > 0$.
(iv) Continuity: $\int |f(x-h) - f(x)|\,dx \to 0$ as $h \to 0$.

Proof. (i) Clear from finite intervals.

(ii) $\int_{-N}^{N} f(x-h)dx = \int_{-N-h}^{N-h} f$, and the end pieces tend to 0 by the decay.

(iii) $\delta \int_{-N}^{N} f(\delta x)dx = \int_{-\delta N}^{\delta N} f$.

(iv) Choose N with $\int_{|x|\ge N} |f(x)|, \int_{|x|\ge N} |f(x-h)| < \epsilon$ for $|h| \le 1$; by uniform continuity on $[-N, N]$, $\int_{-N}^{N} |f(x-h) - f(x)| < \epsilon$ for small h. ∎

5.1.2 Definition of the Fourier transform

For $f \in \mathcal{M}(\mathbb{R})$, $\hat{f}(\xi) = \int_{-\infty}^{\infty} f(x)e^{-2\pi ix\xi}\,dx$ is bounded and continuous. (If $\hat{f} \in \mathcal{M}(\mathbb{R})$ too, inversion and Plancherel hold, §5.1.7.)

5.1.3 The Schwartz space

$\mathcal{S}(\mathbb{R})$ consists of the C^∞ f with $\sup_x |x|^k |f^{(\ell)}(x)| < \infty$ for all $k, \ell \ge 0$ (**rapidly decreasing**); it is closed under differentiation and multiplication by polynomials. It contains the **Gaussian** e^{-ax^2}, $a > 0$ (derivatives $P(x)e^{-ax^2}$), and smooth compactly supported functions such as the **bump function** $e^{-1/(1-x^2)}$ on $|x| < 1$, 0 elsewhere.

5.1.4 The Fourier transform on $\mathcal{S}$

Proposition 1.2

If $f \in \mathcal{S}(\mathbb{R})$, writing $f(x) \to \hat{f}(\xi)$:
(i) $f(x+h) \to \hat{f}(\xi)e^{2\pi ih\xi}$ for $h \in \mathbb{R}$.
(ii) $f(x)e^{-2\pi ixh} \to \hat{f}(\xi+h)$ for $h \in \mathbb{R}$.
(iii) $f(\delta x) \to \delta^{-1}\hat{f}(\delta^{-1}\xi)$ for $\delta > 0$.
(iv) $f'(x) \to 2\pi i\xi\hat{f}(\xi)$.
(v) $-2\pi ixf(x) \to \frac{d}{d\xi}\hat{f}(\xi)$.

Proof. (i)–(iii) by Proposition 1.1(ii), (iii).

(iv) By parts on $[-N,N]$: $\int_{-N}^{N} f'(x)e^{-2\pi ix\xi}dx = [f(x)e^{-2\pi ix\xi}]_{-N}^{N} + 2\pi i\xi \int_{-N}^{N} f(x)e^{-2\pi ix\xi}dx$, with boundary term $\to 0$.

(v) Let $g(x) = -2\pi ixf(x) \in \mathcal{S}$. For $\epsilon > 0$,

$$\frac{\hat{f}(\xi+h)-\hat{f}(\xi)}{h} - \hat{g}(\xi) = \int f(x)e^{-2\pi ix\xi}\Big[\frac{e^{-2\pi ixh}-1}{h} + 2\pi ix\Big]dx.$$

Choose N with $\int_{|x|\geq N}|f|, \int_{|x|\geq N}|x||f| < \epsilon$. For $|x| \leq N$ the bracket is $\leq C|h|$ (Taylor), and always $|\frac{e^{-2\pi ixh}-1}{h}| \leq 2\pi|x|$; so the difference is $\leq C'\epsilon + C''N|h| < C\epsilon$ for small $|h|$. ∎

Theorem 1.3

If $f \in \mathcal{S}(\mathbb{R})$, then $\hat{f} \in \mathcal{S}(\mathbb{R})$.

Proof. $\hat{f}$ is bounded. By Proposition 1.2(iv), (v), the function $\xi^k(\frac{d}{d\xi})^\ell \hat{f}$ is the transform of $\frac{1}{(2\pi i)^k}(\frac{d}{dx})^k[(-2\pi ix)^\ell f] \in \mathcal{S}$, hence bounded; $\hat{f} \in C^\infty$ by repeated (v). ∎

Theorem 1.4

If $f(x) = e^{-\pi x^2}$, then $\hat{f}(\xi) = f(\xi)$.

Proof. $F = \hat{f}$ has $F(0) = \int e^{-\pi x^2}dx = 1$. By (v), $f' = -2\pi xf$ and (iv),

$$F'(\xi) = \int f(x)(-2\pi ix)e^{-2\pi ix\xi}dx = i\int f'(x)e^{-2\pi ix\xi}dx = i(2\pi i\xi)F(\xi) = -2\pi\xi F(\xi).$$

So $G = Fe^{\pi\xi^2}$ has $G' = 0$, $G \equiv 1$. ∎

($\int e^{-\pi x^2}dx = 1$: square it and use polar coordinates.)

Corollary 1.5

If $\delta > 0$ and $K_\delta(x) = \delta^{-1/2}e^{-\pi x^2/\delta}$, then $\widehat{K_\delta}(\xi) = e^{-\pi\delta\xi^2}$.

Proof. Proposition 1.2(iii) with δ replaced by $\delta^{-1/2}$ and Theorem 1.4. ∎

As $\delta \to 0$, K_δ concentrates while $\widehat{K_\delta}$ flattens: the uncertainty principle in miniature (§5.4).

Definition (Good kernels on $\mathbb{R}$)

$\{K_\delta\}_{\delta>0}$ is a family of **good kernels** as $\delta \to 0$ if (i) $\int K_\delta = 1$;
(ii) $\int |K_\delta| \leq M$;
(iii) for every $\eta > 0$, $\int_{|x|>\eta}|K_\delta| \to 0$.

Theorem 1.6

$\{K_\delta\}_{\delta>0}$, $K_\delta(x) = \delta^{-1/2}e^{-\pi x^2/\delta}$, is a family of good kernels as $\delta \to 0$.

Proof. $K_\delta > 0$, $\int K_\delta = \widehat{K_\delta}(0) = 1$, and with $x = \delta^{1/2}y$, $\int_{|x|>\eta} K_\delta = \int_{|y|>\eta/\sqrt{\delta}} e^{-\pi y^2}dy \to 0$. ∎

For $f, g \in \mathcal{S}(\mathbb{R})$, $(f*g)(x) = \int f(x-t)g(t)\,dt$.

Corollary 1.7

If $f \in \mathcal{S}(\mathbb{R})$, then $(f * K_\delta)(x) \to f(x)$ uniformly in x as $\delta \to 0$.

Proof. f is uniformly continuous ($|f| < \epsilon/4$ for $|x| \geq R$, uniform continuity on $[-R-1, R+1]$); take η with $|f(x-t) - f(x)| < \epsilon$ for $|t| < \eta$. Then

$$|(f * K_\delta)(x) - f(x)| \leq \int_{|t|<\eta} K_\delta(t)|f(x-t) - f(x)|dt + \int_{|t|\geq\eta} K_\delta(t)\, 2\sup|f|\, dt \leq \epsilon + C\int_{|t|\geq\eta} K_\delta < 2\epsilon$$

for δ small, uniformly in x. ■

5.1.5 The Fourier inversion

Proposition 1.8 (Multiplication formula)

If $f, g \in \mathcal{S}(\mathbb{R})$, then $\int f(x)\hat{g}(x)\, dx = \int \hat{f}(y)g(y)\, dy$.

Proof. $F(x,y) = f(x)g(y)e^{-2\pi ixy}$ has $|F| \leq \frac{C}{(1+x^2)(1+y^2)}$, so by Theorem 3.1 the iterated integrals, $\int f\hat{g}$ and $\int \hat{f}g$, agree. ■

Theorem 1.9 (Fourier inversion)

If $f \in \mathcal{S}(\mathbb{R})$, then $f(x) = \int_{-\infty}^{\infty} \hat{f}(\xi)e^{2\pi ix\xi}\, d\xi$.

Proof.

Step 1 ($x = 0$). $G_\delta(x) = e^{-\pi\delta x^2}$ has $\widehat{G_\delta} = K_\delta$, so $\int fK_\delta = \int \hat{f}G_\delta$ (Proposition 1.8). As $\delta \to 0$ the left side tends to $f(0)$ (Corollary 1.7, K_δ even) and the right to $\int \hat{f}$ ($G_\delta \to 1$ uniformly on compacts, dominated by $|\hat{f}|$).

Step 2 (general x). Apply Step 1 to $f(\cdot + x)$, with transform $\hat{f}(\xi)e^{2\pi ix\xi}$. ■

With $\mathcal{F}(f) = \hat{f}$ and $\mathcal{F}^*(g)(x) = \int g(\xi)e^{2\pi ix\xi}d\xi = \mathcal{F}(g)(-x)$, $\mathcal{F}^*\mathcal{F} = I$ on $\mathcal{S}$.

Corollary 1.10

The Fourier transform is a bijection of $\mathcal{S}(\mathbb{R})$ onto itself.

Proof. $\mathcal{F}(\mathcal{S}) \subseteq \mathcal{S}$ (Theorem 1.3), $\mathcal{F}^*\mathcal{F} = I$ (Theorem 1.9), and $\mathcal{F}\mathcal{F}^* = I$ as $\mathcal{F}^*(g) = \mathcal{F}(g)(-\cdot)$. ■

5.1.6 The Plancherel formula

Proposition 1.11

If $f, g \in \mathcal{S}(\mathbb{R})$:
(i) $f * g \in \mathcal{S}(\mathbb{R})$;
(ii) $f * g = g * f$;
(iii) $\widehat{f * g}(\xi) = \hat{f}(\xi)\hat{g}(\xi)$.

Proof.

(ii) Substitute $t \mapsto x - t$.

(i) $|x|^\ell \le 2^\ell(|x-y|^\ell+|y|^\ell)$ gives $\sup_x |x|^\ell|g(x-y)| \le A_\ell(1+|y|)^\ell$, so $\sup_x |x^\ell(f*g)| \le A_\ell \int |f(y)|(1+|y|)^\ell dy < \infty$; and $\frac{d}{dx}(f*g) = f*g'$, repeated.

(iii) $f(y)g(x-y)e^{-2\pi ix\xi}$ is integrable on $\mathbb{R}^2$, so by Fubini (Theorem 3.1), $\int (f*g)(x)e^{-2\pi ix\xi}dx = \int f(y)e^{-2\pi iy\xi}\Big(\int g(x-y)e^{-2\pi i(x-y)\xi}dx\Big)dy = \hat f(\xi)\hat g(\xi)$. ■

Theorem 1.12 (Plancherel)

If $f \in \mathcal{S}(\mathbb{R})$, then $\|\hat f\| = \|f\|$, where $\|f\| = \big(\int |f|^2\big)^{1/2}$.

Proof. $f^\flat(x) = \overline{f(-x)}$ has $\widehat{f^\flat} = \overline{\hat f}$, so $h = f*f^\flat$ has $\hat h = |\hat f|^2$ and $h(0) = \int |f|^2$; inversion at 0 gives $h(0) = \int |\hat f|^2$. ■

5.1.7 Extension to functions of moderate decrease

For $f, g \in \mathcal{M}(\mathbb{R})$, $f*g \in \mathcal{M}(\mathbb{R})$, $\widehat{f*g} = \hat f\hat g$, and the multiplication formula holds; so inversion and Plancherel hold whenever $f, \hat f \in \mathcal{M}(\mathbb{R})$.

5.1.8 The Weierstrass approximation theorem

Theorem 1.13 (Weierstrass)

If f is continuous on $[a,b]$ and $\epsilon > 0$, there is a polynomial P with $\sup_{[a,b]} |f-P| < \epsilon$.

Proof. Take $[-M,M] \supset [a,b]$ in its interior and extend f to continuous g on $\mathbb{R}$, linear to 0 on $[b,M]$ and $[-M,a]$, 0 outside; $|g| \le B$. As in Corollary 1.7, fix δ with $|g*K_\delta - g| < \epsilon/2$. Truncating the Taylor series of $e^{-\pi x^2/\delta}$ (uniform on $|x| \le 2M$) gives a polynomial R with $|K_\delta - R| < \frac{\epsilon}{4MB}$ on $|x| \le 2M$. Then $P(x) = \int_{-M}^{M} g(t)R(x-t)\,dt$ is a polynomial, and for $|x| \le M$,

$$|(g*K_\delta)(x) - P(x)| = \Big|\int_{-M}^{M} g(t)(K_\delta(x-t) - R(x-t))dt\Big| \le 2MB\frac{\epsilon}{4MB} = \frac{\epsilon}{2}.$$

■

5.2 Applications to some partial differential equations

Motivation. ∂_x^2 becomes multiplication by $-4\pi^2\xi^2$, so the heat and Laplace equations are solved by Gaussian and Poisson convolutions; uniqueness needs control at infinity.

5.2.1 The time-dependent heat equation on the real line

Seek u with $\partial_t u = \partial_x^2 u$ $(t > 0)$, $u(x,0) = f(x)$. In x: $\partial_t \hat u = -4\pi^2\xi^2\hat u$, so $\hat u = \hat f(\xi)e^{-4\pi^2\xi^2 t}$ and $u = f * \mathcal{H}_t$, the **heat kernel** being

$$\mathcal{H}_t(x) = K_\delta(x) \text{ with } \delta = 4\pi t, \qquad \mathcal{H}_t(x) = \frac{1}{(4\pi t)^{1/2}}e^{-x^2/4t}, \qquad \widehat{\mathcal{H}_t}(\xi) = e^{-4\pi^2 t\xi^2}.$$

Theorem 2.1

Given $f \in \mathcal{S}(\mathbb{R})$, let $u(x,t) = (f * \mathcal{H}_t)(x)$ for $t > 0$. Then:
(i) u is C^2 for $x \in \mathbb{R}$, $t > 0$, and solves the heat equation.
(ii) $u(x,t) \to f(x)$ uniformly in x as $t \to 0$; hence, setting $u(x,0) = f(x)$, u is continuous on $\{t \geq 0\}$.
(iii) $\int |u(x,t) - f(x)|^2 dx \to 0$ as $t \to 0$.

Proof.

(i) $u = \int \hat{f}(\xi) e^{-4\pi^2\xi^2 t} e^{2\pi i x\xi} d\xi$ (Proposition 1.11 and inversion); differentiating under the integral (rapid decay of $\hat{f}$) gives $\partial_t u = \partial_x^2 u$.

(ii) Corollary 1.7 with $\delta = 4\pi t$.

(iii) By Plancherel, $\int |u - f|^2 = \int |\hat{f}|^2 |e^{-4\pi^2\xi^2 t} - 1|^2 d\xi$; split at $|\xi| = N$, using $|e^{-4\pi^2\xi^2 t} - 1| \leq 4\pi^2 N^2 t$ inside and ≤ 1 with $\int_{|\xi|>N} |\hat{f}|^2 < \epsilon$ outside. ∎

Corollary 2.2

$u(\cdot, t) \in \mathcal{S}(\mathbb{R})$ uniformly in t: for every $T > 0$ and $k, \ell \geq 0$,

$$\sup_{x \in \mathbb{R},\ 0<t<T} |x|^k |\partial_x^\ell u(x,t)| < \infty. \tag{5.1}$$

Proof. $\partial_x^\ell u = (\partial_x^\ell f) * \mathcal{H}_t$, so take $\ell = 0$, $|x| \geq 1$, and split $\int f(x-y)\mathcal{H}_t(y)dy$ at $|y| = |x|/2$. For $|y| \leq |x|/2$, $|f(x-y)| \leq C_k |x|^{-k}$ and $\int \mathcal{H}_t = 1$. For $|y| \geq |x|/2$, $\mathcal{H}_t(y) \leq (4\pi t)^{-1/2} e^{-x^2/16t}$, and [Supplement] with $s = x^2/t \geq x^2/T$, $t^{-1/2} e^{-x^2/16t} = |x|^{-1} s^{1/2} e^{-s/16} \leq C|x|^{-1} e^{-x^2/32T} \leq C_{k,T} |x|^{-k}$; with $\int |f| < \infty$, $|u(x,t)| \leq C|x|^{-k}$. ∎

Theorem 2.3 (Uniqueness)

Suppose (i) u is continuous on $\{t \geq 0\}$;
(ii) u satisfies the heat equation for $t > 0$;
(iii) $u(x,0) = 0$;
(iv) $u(\cdot, t) \in \mathcal{S}(\mathbb{R})$ uniformly in t as in (5.1). Then $u = 0$.

Proof. $E(t) = \int |u(x,t)|^2 dx \geq 0$, $E(0) = 0$. Differentiating and integrating by parts (decay kills boundary terms),

$$\frac{dE}{dt} = \int (\partial_t u\, \bar{u} + u\, \overline{\partial_t u}) = \int (\partial_x^2 u\, \bar{u} + u\, \overline{\partial_x^2 u}) = -2 \int |\partial_x u|^2 \leq 0.$$

So E decreases, $E \equiv 0$, and $u = 0$. ∎

(Without decay conditions such as (iv), uniqueness fails.)

5.2.2 The steady-state heat equation in the upper half-plane

Find u on $\mathbb{R}^2_+ = \{y > 0\}$ with $\triangle u = 0$, $u(x,0) = f(x)$. In x: $\partial_y^2 \hat{u} = 4\pi^2\xi^2 \hat{u}$, so $\hat{u} = Ae^{2\pi|\xi|y} + Be^{-2\pi|\xi|y}$; boundedness gives $\hat{u} = \hat{f}(\xi) e^{-2\pi|\xi|y}$ and $u = f * \mathcal{P}_y$, the **Poisson kernel** $\mathcal{P}_y(x) = \frac{1}{\pi} \frac{y}{x^2+y^2}$.

Lemma 2.4

$\int_{-\infty}^{\infty} e^{-2\pi|\xi|y} e^{2\pi i\xi x} d\xi = \mathcal{P}_y(x)$ and $\int_{-\infty}^{\infty} \mathcal{P}_y(x) e^{-2\pi i x\xi} dx = e^{-2\pi|\xi|y}$.

Proof. Splitting at 0: $\int_0^\infty e^{-2\pi\xi(y-ix)}d\xi + \int_0^\infty e^{-2\pi\xi(y+ix)}d\xi = \frac{1}{2\pi(y-ix)} + \frac{1}{2\pi(y+ix)} = \frac{1}{\pi}\frac{y}{x^2+y^2}$. Both functions are of moderate decrease, so inversion (§5.1.7) gives the second identity. ■

Lemma 2.5

$\{\mathcal{P}_y\}$ is a family of good kernels on $\mathbb{R}$ as $y \to 0$.

Proof. $\mathcal{P}_y > 0$; $\int \mathcal{P}_y = \widehat{\mathcal{P}_y}(0) = 1$ (or $\frac{1}{\pi}\int \frac{dx}{1+x^2} = 1$ after $x \mapsto yx$); $\int_{|x|\geq\delta} \mathcal{P}_y = \frac{1}{\pi}\int_{|x|\geq\delta/y} \frac{dx}{1+x^2} \to 0$. ■

Theorem 2.6

Given $f \in \mathcal{S}(\mathbb{R})$, let $u(x,y) = (f * \mathcal{P}_y)(x)$. Then:
(i) u is C^2 in $\mathbb{R}^2_+$ and $\triangle u = 0$;
(ii) $u(x,y) \to f(x)$ uniformly as $y \to 0$;
(iii) $\int |u(x,y) - f(x)|^2 dx \to 0$ as $y \to 0$;
(iv) with $u(x,0) = f(x)$, u is continuous on $\overline{\mathbb{R}^2_+}$ and vanishes at infinity: $u(x,y) \to 0$ as $|x| + y \to \infty$.

Proof.

(i)–(iii) As for Theorem 2.1: $u = \int \hat{f}(\xi) e^{-2\pi|\xi|y} e^{2\pi i x\xi} d\xi$ and $(\partial_x^2 + \partial_y^2) e^{-2\pi|\xi|y} e^{2\pi i x\xi} = 0$; Lemma 2.5; Plancherel.

(iv) Splitting at $|t| = |x|/2$, $|u| \leq C(\frac{1}{1+x^2} + \frac{y}{x^2+y^2})$; also $|u| \leq \sup \mathcal{P}_y \int |f| \leq C/y$. Use the first for $|x| \geq y$, the second for $|x| \leq y$. ■

Theorem 2.7

If u is continuous on $\overline{\mathbb{R}^2_+}$, $\triangle u = 0$ in $\mathbb{R}^2_+$, $u(x,0) = 0$, and u vanishes at infinity, then $u = 0$.

Lemma 2.8 (Mean-value property)

Let u be C^2 with $\triangle u = 0$ in an open $\Omega \subseteq \mathbb{R}^2$. If the closed disc of radius R centred at (x,y) lies in Ω, then $u(x,y) = \frac{1}{2\pi}\int_0^{2\pi} u(x + r\cos\theta, y + r\sin\theta)\, d\theta$ for $0 \leq r \leq R$.

Proof. $U(r,\theta) = u(x + r\cos\theta, y + r\sin\theta)$ satisfies $0 = \partial_\theta^2 U + r\partial_r(r\partial_r U)$. For $F(r) = \frac{1}{2\pi}\int_0^{2\pi} U\, d\theta$, $r\partial_r(r\partial_r F) = -\frac{1}{2\pi}\int \partial_\theta^2 U = 0$, so rF' is constant, $= 0$ at $r = 0$, and $F(r) = F(0) = u(x,y)$. ■

Proof of Theorem 2.7. ($u = y$ shows the decay is needed.) Take u real and suppose $u(x_0,y_0) > 0$. Choose R with $u \leq \frac{1}{2}u(x_0,y_0)$ outside $D_R^+ = \{x^2 + y^2 \leq R^2,\ y \geq 0\}$; the maximum M of u on D_R^+, at (x_1,y_1), is its maximum on $\overline{\mathbb{R}^2_+}$, and $y_1 > 0$. By the mean-value property for $r < y_1$, $M = \frac{1}{2\pi}\int u(x_1 + r\cos\theta, y_1 + r\sin\theta)d\theta \leq M$, so $u = M$ on these circles, hence on the disc of radius y_1, whose boundary meets $y = 0$ where $u = 0$: a contradiction. Similarly $u \geq 0$. ■

5.3 The Poisson summation formula

Motivation. Periodizing f gives a function on the circle with Fourier coefficients $\hat{f}(n)$, linking $\mathbb{R}$ and the circle: the theta functional equation and the circle's heat and Poisson kernels.

For $f \in \mathcal{S}(\mathbb{R})$ the **periodization** $F_1(x) = \sum_{n\in\mathbb{Z}} f(x+n)$ is 1-periodic and continuous; compare $F_2(x) = \sum_n \hat{f}(n) e^{2\pi i n x}$.

Theorem 3.1 (Poisson summation formula)

If $f \in \mathcal{S}(\mathbb{R})$, then $\sum_{n\in\mathbb{Z}} f(x+n) = \sum_{n\in\mathbb{Z}} \hat{f}(n)e^{2\pi inx}$. In particular $\sum_n f(n) = \sum_n \hat{f}(n)$.

Proof. Both sides are continuous and 1-periodic, the right absolutely convergent. The m-th coefficient of the left side is

$$\int_0^1 \sum_n f(x+n)e^{-2\pi imx}dx = \sum_n \int_n^{n+1} f(y)e^{-2\pi imy}dy = \int_{-\infty}^{\infty} f(y)e^{-2\pi imy}dy = \hat{f}(m),$$

by uniform convergence, as for the right side; Corollary 2.2 (period 1) gives equality. ■

(Likewise when $f, \hat{f} \in \mathcal{M}(\mathbb{R})$.)

5.3.1 Theta and zeta functions

The **theta function** is $\vartheta(s) = \sum_{n\in\mathbb{Z}} e^{-\pi n^2 s}$ for $s > 0$.

Theorem 3.2

$s^{-1/2}\vartheta(1/s) = \vartheta(s)$ for $s > 0$.

Proof. Poisson summation at $x = 0$ for $f = e^{-\pi sx^2}$, $\hat{f}(\xi) = s^{-1/2}e^{-\pi\xi^2/s}$ (Corollary 1.5, $\delta = 1/s$). ■

Via $\pi^{-s/2}\Gamma(s/2)\zeta(s) = \int_0^\infty t^{s/2-1}\frac{\vartheta(t)-1}{2}dt$ this gives the functional equation of $\zeta(s) = \sum n^{-s}$ (Book II).

5.3.2 Heat kernels

Theorem 3.3

The heat kernel of the circle is the periodization of the heat kernel of the line: $H_t(x) = \sum_{n\in\mathbb{Z}} \mathcal{H}_t(x+n)$.

Proof. Poisson summation for $f = \mathcal{H}_t$, $\hat{f}(\xi) = e^{-4\pi^2 t\xi^2}$: $\sum_n \mathcal{H}_t(x+n) = \sum_n e^{-4\pi^2 tn^2}e^{2\pi inx} = H_t(x)$. ■

Corollary 3.4

H_t is a good kernel as $t \to 0$ (on the circle $\mathbb{R}/\mathbb{Z}$, with integrals over $|x| \le \frac{1}{2}$).

Proof. $H_t > 0$ by Theorem 3.3, $\int_{|x|\le 1/2} H_t = \hat{H}_t(0) = 1$. For $|x| \le \frac{1}{2}$, $H_t = \mathcal{H}_t + E_t$ with $E_t(x) = \sum_{n\neq 0} \mathcal{H}_t(x+n) \le Ct^{-1/2}\sum_{n\ge 1} e^{-cn^2/t} \to 0$ uniformly ($|x+n| \ge |n|/2$), and $\mathcal{H}_t(x) \le (4\pi t)^{-1/2}e^{-\delta^2/4t} \to 0$ for $\delta \le |x| \le \frac{1}{2}$. ■

5.3.3 Poisson kernels

Theorem 3.5

$P_r(2\pi x) = \sum_{n\in\mathbb{Z}} \mathcal{P}_y(x+n)$ with $r = e^{-2\pi y}$.

Proof. Poisson summation for $f = \mathcal{P}_y$, $\hat{f} = e^{-2\pi|\xi|y}$ (moderate decrease): $\sum_n \mathcal{P}_y(x+n) = \sum r^{|n|} e^{in(2\pi x)} = P_r(2\pi x)$. ∎

5.4 The Heisenberg uncertainty principle

Motivation. f and $\hat{f}$ cannot both be sharply localized: the product of their spreads is at least $1/16\pi^2$, with equality only for Gaussians.

Theorem 4.1 (Heisenberg)

If $\psi \in \mathcal{S}(\mathbb{R})$ with $\int |\psi|^2 = 1$, then

$$\Big(\int x^2|\psi(x)|^2dx\Big)\Big(\int \xi^2|\hat{\psi}(\xi)|^2d\xi\Big) \geq \frac{1}{16\pi^2},$$

with equality iff $\psi(x) = Ae^{-Bx^2}$ with $B > 0$ and $|A|^2 = \sqrt{2B/\pi}$. More generally $\big(\int (x-x_0)^2|\psi|^2\big)\big(\int(\xi-\xi_0)^2|\hat{\psi}|^2\big) \geq \frac{1}{16\pi^2}$ for all $x_0, \xi_0 \in \mathbb{R}$.

Proof.

Step 1. By parts (ψ, ψ' rapidly decreasing),

$$1 = \int |\psi|^2 = -\int x\frac{d}{dx}|\psi|^2 = -\int (x\psi'\bar{\psi} + x\psi\overline{\psi'}),$$

so by Cauchy–Schwarz $1 \leq 2\int |x||\psi||\psi'| \leq 2\big(\int x^2|\psi|^2\big)^{1/2}\big(\int |\psi'|^2\big)^{1/2}$.

Step 2. By Proposition 1.2(iv) and Plancherel, $\int |\psi'|^2 = 4\pi^2 \int \xi^2|\hat{\psi}|^2$. Squaring gives the inequality.

Step 3 (equality). Equality in Cauchy–Schwarz forces $\psi' = \beta x\psi$, so $\psi = Ae^{\beta x^2/2}$; $\psi \in \mathcal{S}$ forces $\beta = -2B < 0$, and $\int |\psi|^2 = |A|^2\sqrt{\pi/2B} = 1$.

Step 4. For the general form apply the inequality to $e^{-2\pi i x\xi_0}\psi(x+x_0)$. ∎

Quantum mechanics. For a state ψ, $\int |\psi|^2 = 1$, the particle lies in (a,b) with probability $\int_a^b |\psi|^2$, with mean $\bar{x} = \int x|\psi|^2$ and variance $(\Delta x)^2 = \int (x-\bar{x})^2|\psi|^2$. Momentum is governed likewise by $\hat{\psi}$ (frequency ξ corresponds to momentum $h\xi$), and Theorem 4.1 reads $(\Delta x)(\Delta\xi) \geq \frac{1}{4\pi}$, i.e. $(\Delta\,\text{position})(\Delta\,\text{momentum}) \geq \frac{h}{4\pi}$.